Exercise 2

Database : employees_db

employees

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	45 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

1. Count () function

Write a SQL query to find the total number of employees in the company

```
SELECT    COUNT(*)  AS total_employees
FROM      employees;
```

total_employees
10

2. SUM() Function

Write a SQL Query to find the total salary paid to all employees in the IT department

```
SELECT      SUM(salary) AS total_salary,
            department
FROM        employees
WHERE       department = 'IT'
GROUP BY    department;
```

total_salary	department
220 000	IT

3. AVG() Function

Write a SQL query to calculate the average salary of employees in the HR department.

```
SELECT      AVG(salary) AS average_salary,
            department
FROM        employees
WHERE       department = 'HR'
GROUP BY    department;
```

average_salary	department
49 500	HR

5. GROUP BY statement

Write a SQL query to group employees by department & display the total salary paid in each department.

```
SELECT      SUM(salary) AS total_salary,
            department
FROM        employees
GROUP BY    department;
```

total_salary	department
220 000	IT
99 000	HR
119 000	Finance
105 000	Marketing

6. GROUP BY and COUNT()

Write a SQL query to count how many employees work in each city

```
SELECT      COUNT(*)      AS number_of_employees,
            city
FROM        employees
GROUP BY    city;
```

number_of_employees	city
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7. GROUP BY and ORDER BY

Write a SQL query to group employees by department, calculate average salary in each department, and order the results in descending order of average salary

```
SELECT      department,
            AVG(salary) AS average_salary
FROM        employees
GROUP BY    department
ORDER BY    AVG(salary) DESC;
```

department	average salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

8. HAVING CLAUSE

Write a SQL query to find departments where the total salary paid exceeds 100 000 (use GROUP BY and HAVING)

```
SELECT      department,
            SUM (salary) AS total_salary
FROM        employees
GROUP BY    department
HAVING      SUM (salary) > 100000;
```

department	total_salary
IT	220 000
Finance	119 000
Marketing	105000

9. Combining GROUP BY, HAVING and ORDER BY

Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

```
SELECT      COUNT (*) AS number_of_employees,
            city
FROM        employees
GROUP BY    city
HAVING      COUNT(*) > 1
ORDER BY    COUNT(*) DESC;
```

number_of_employees	city
3	Chicago
2	New York
2	San Francisco

16. <u>Combining Aggregate Functions</u>

Write a SQL query to find the department with the highest average salary

```
SELECT    AVG (salary) AS average_salary,
          department
FROM      employees
GROUP BY  department
ORDER BY  average_salary DESC
LIMIT     1 ;
```

average_salary	department
59500	Finance

4. <u>MIN () and MAX () functions</u>

Write a SQL query to find the highest & lowest salary in the company

```
SELECT    MAX (salary) AS highest_salary,
          MIN (salary) AS lowest_salary
FROM      employees ;
```

highest salary	lowest_salary
62 000	48000